KK-L-241D
July 31, 1969
SUPERSEDING
Fed. Spec. KK-L-241c
July 1, 1964

FEDERAL SPECIFICATION

LEATHER, CATTLEHIDE, RIGGING

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers rigging leather made from full grain cattlehide.

1.2 Classification.

1.2.1 Types and classes. The rigging leather shall be furnished in the following types and classes, as specified (see 6.2):

Type I – Sides.

Type II – Backs.

Class 1 – Mildew resistant treated.

Class 2 – Not mildew resistant treated.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 311 - Leather, Methods of Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from the Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-L-43432 - Leather, Packaging and Packing of.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Material. The leather shall be full grain cattlehide. Bull hides will not be acceptable. The flesh side of the leather shall be smooth and free of coarse loose flesh.

3.2 Tannage. The leather shall be vegetable tanned.

3.3 Physical requirements.

3.3.1 Trim (form and area). The leather shall be furnished in the form of sides or backs as required for the type specified (see 6.2) and shall be trimmed to conform with the definition of sides or backs as listed under "Definition of Terms" in Fed. Test Method Std. No. 311. The area in terms of square feet shall be legibly marked in the butt area on the flesh side of the leather.

3.3.2 Color. Unless otherwise specified (see 6.2), the color of the leather shall be natural as produced by the tanning agents. If necessary, a light application of dye or pigment may be employed to level the color of the grain side.

3.3.3 Weight and thickness. The leather shall be of the weight and thickness specified (see 6.2) and as shown in table I. The leather may be shaved on the flesh side to provide greater uniformity of thickness, but shall not be split (see 6.4).

TABLE I. Weight and thickness

Weight	Thickness (ounces) 1/
Light	6 to 7-3/4
Medium	8 to 9-3/4
Heavy	10 and above

1/ 1 ounce = 1/64 inch.

3.3.4 Breaking force. Not less than 80 percent of the specimens tested shall meet the minimum breaking force values indicated in table II when tested as specified in 4.3.

TABLE II. Breaking force

Thickness (ounces)	Breaking force minimum (pounds)
6 to 7-3/4	150
8 to 9-3/4	200
10 and above	250

3.3.5 Elongation. Not less than 80 percent of the specimens tested shall have elongation values of 20 percent or less, at the loads shown in table III when tested as specified in 4.3.

TABLE III. Elongation load

Thickness (ounces)	Elongation load (pounds)
6 to 7-3/4	100
8 to 9-3/4	150
10 and above	200

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3.3.6 Cracking. The leather shall not crack when tested as specified in 4.3.

3.4 Chemical requirements.

3.4.1 Class 1. When class 1 leather is specified (see 6.2), it shall contain not less than 0.18 percent nor more than 0.70 percent paranitrophenol fungicide when tested as specified in 4.3.

3.4.2 The leather shall conform to the requirements of table IV when tested as specified in 4.3.

TABLE IV. Chemical requirements

Characteristic	Minimum	Maximum
Chloroform-soluble material (percent) 1/	10.0	18.0
Water-soluble material (percent) 1/	-	18.0
Total ash (percent) 1/	-	5.0
Degree of tannage	45.0	70.0
pH value	3.3	4.5

1/ Calculated on the moisture free basis.

3.5 Workmanship. The leather shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Inspection. Sampling for inspection shall be performed in accordance with Fed. Test Method Std. No. 311, except when otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1, above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.2 Examination of the end item. The defects found during examination of the leather shall be classified in accordance with 4.2.2.1 and 4.2.2.2. The sample unit for these examinations shall be one side or back. The lot size shall be expressed in units of one side or back. Sampling for these examinations shall be performed in accordance with Fed. Test Method Std. No. 311 except that a sample size of 15 and a rejection number of 2 shall be applicable for thickness determinations on lots consisting of 91 or more units of products.

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4.2.2.1 Examination for visual characteristics.

Examine	Defect
Material	Not full grain cattlehide.
Trim	Not sides or not backs, as required.
Color	Unless otherwise specified, color not natural.
Quality of leather	Flesh side not smooth or contains areas of loose flesh in excess of 4 square inches in total area. Any hole, deep cut, scratch, damaged grain, hipmark, rough shoulder, stain, or other defects which impair the cutting value of more than 10 percent of the total area of the back or side.

NOTE: Light well healed scratches or grub holes, slight wrinkles, or stains shall not be classified as defects.

4.2.2.2 Examination for dimensional characteristcs.

Examine	Defect
Area (expressed in terms of square feet)	Less than area marked on leather.
Marking	Omitted, illegible or not in area specified.
Thickness 1/	Not as specified.

1/ Thickness shall be determined in accordance with method 1011 of **Fed. Test Method Std. No. 311.**

4.2.3 Examination for preparation for delivery requirements. An examination shall be made in accordance with the provisions of MIL-L-43432 to determine that packaging, packing and marking requirements of section 5 of this specification are complied with.

4.3 End item testing. **The methods of testing specified in Fed. Test Method Std. No. 311, wherever applicable and as listed in table V, shall be followed. Sampling procedure and location from which the sample unit is to be obtained shall be in accordance with Fed. Test Method Std. No. 311. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be rejected if any one of the following conditions exist:**

a. More than three test failures occur for breaking force or elongation.
b. More than one test failure occurs for cracking.
c Any composite test result fails to meet the specified requirement.

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TABLE V. End item tests

Characteristic	Requirement paragraph	Test method	Requirement applicable to: Sample unit	Composite
Material	3.1	1/	-	-
Tannage	3.2	1/	-	-
Breaking force	3.3.4	2021	X	-
Elongation	3.3.5	2021	X	-
Cracking	3.3.6	4.3.1	X	-
Paranitrophenol	3.4.1	6711	-	X
Chloroform-soluble material	3.4.2	6341	-	X
Water-soluble material	3.4.2	6341	-	X
Total ash	3.4.2	6421	-	X
Degree of tannage	3.4.2	6631	-	X
pH value	3.4.2	6621	-	X

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the requirement stated.

4.3.1 Cracking. The leather shall be tested for cracking in accordance with method 4011 of Fed. Test Method Std. No. 311. The diameter of the mandrel used for testing the leather shall be as follows:

Thickness of leather (ounces)	Diameter of mandrel (inch)
6 to 7-3/4	1/4
8 to 9-3/4	1/4
10 and above	5/16

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Levels A and C. Leather shall be packaged in accordance with the applicable requirements of MIL-L-43432.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Levels A, B, or C. Leather shall be packed in accordance with the applicable requirements of MIL-L-43432.

5.3 Marking.

5.3.1 Civil requirements. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std, No. 123.

5.3.2 Military requirements. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-L-43432.

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6. NOTES

6.1 Intended use. The leather is intended for use in shoe and textile repair shops, trailer mounted.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

(a) Title, number and date of this specification.
(b) Type and class required (see 1.2.1).
(c) Color when other than natural color is required (see 3.3.2).
(d) Weight and thickness required (see 3.3.3).
(e) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Civil agency procurement. When level B packaging is specified for civil agency procurement, the level A requirements in 5.1.1 shall apply.

6.4 Strength and elongation characteristics are known to be adversely affected when thick layers of leather are removed from the flesh side by splitting.

MILITARY CUSTODIANS:

Army - GL
Navy - SA

Review activities:

Army - MD
Navy - SA

Preparing activity:

Army - GL

CIVIL AGENCIES INTEREST::

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side; staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
KK-L-241D; Leather, Cattlehide, Rigging

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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